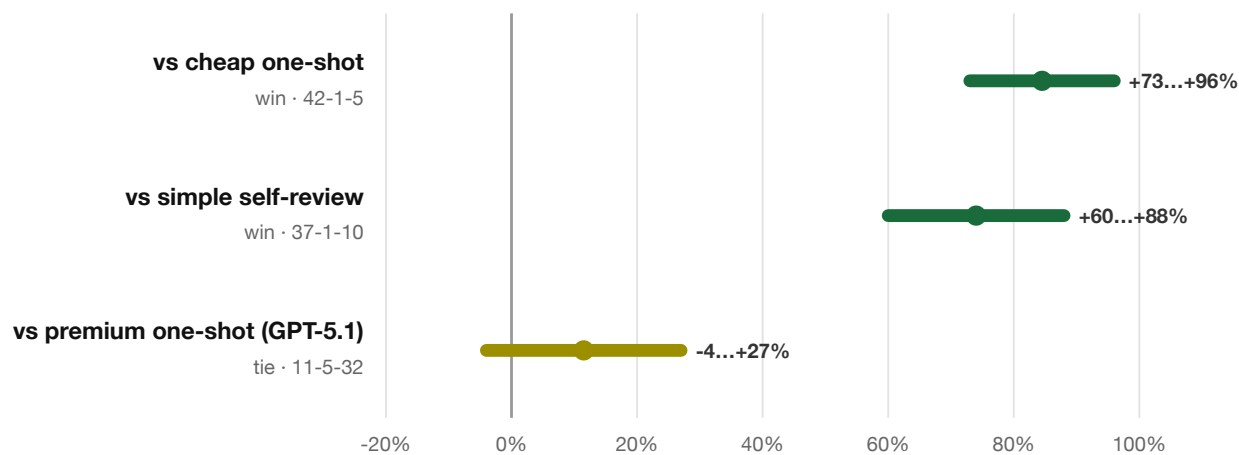


# Depth beats breadth: cheap model + adversarial review

Frugal Fusion experiment — 48 hard engineering tasks, 3-model blind judge panel, 95% bootstrap CIs

A fresh-eyes adversarial review loop (draft → multi-lens critics → skeptic filter → revise, to convergence) run on a **single cheap model** (qwen3-235b) reaches **premium-model (GPT-5.1) quality parity at 0.66× the cost** on hard engineering tasks, and decisively beats its own single-shot and simple self-review baselines. A separate multi-model ensemble ("fusion") approach was tested first and did **not** beat a single strong cheap model — it lost on both deterministic and open-ended tasks.

## Net win-rate vs. baseline (95% bootstrap CI)



Dot = point estimate, bar = 95% CI. CI crossing 0% = statistical tie. Record format is win-tie-loss out of 48 tasks.

| Comparison                            | Net win-rate 95% CI                                         | Record (W-T-L) | Verdict         |
|---------------------------------------|-------------------------------------------------------------|----------------|-----------------|
| Review vs. cheap one-shot             | +73% to +96%                                                | 42-1-5         | Significant win |
| Review vs. simple self-review         | +60% to +88%                                                | 37-1-10        | Significant win |
| Review vs. premium one-shot (GPT-5.1) | −4% to +27%                                                 | 11-5-32        | Statistical tie |
| Cost (review vs. premium one-shot)    | 0.66× — answer lengths comparable, not a verbosity artifact |                |                 |

**What is and isn't claimed:** depth (adversarial review on one cheap model) reaches premium-quality parity at ~2/3 cost on hard tasks, and massively beats single-shot / simple self-review. It is **not** claimed that review beats premium (the CI crosses zero) — nor that multi-model fusion helps (it didn't) — nor that this holds on easy tasks (no headroom to show a difference) or across all future model generations.

Method, every experimental round, and raw numbers: docs/EXPERIMENT\_RESULTS.md and docs/PUBLICATION.md in the source repository.